An IoT-Inspired Climate-Resilient and Explainable Crop Recommendation System Using Machine Learning

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***Abstract*—Smart agriculture demands data-driven decision support to address the challenges of climate variability and soil management. This paper presents a crop recommendation system designed for IoT-based deployment, leveraging soil nutrient and environmental sensor data. Four classifiers—Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), and Gradient Boosting (GB)—are evaluated on a dataset of 7,630 samples spanning 12 Ethiopian crop classes and 14 agro-climatic features. Class imbalance is addressed using the Synthetic Minority Oversampling Technique (SMOTE), yielding a balanced training corpus of 10,716 samples. Random Forest achieves the highest accuracy of 84.33%. Explainability is provided through SHAP (SHapley Additive exPlanations) analysis, identifying potassium (K), sulfur (S), and phosphorus (P) as the three most influential features. As the primary novel contribution, climate resilience is evaluated by systematically perturbing temperature ($\pm 2^\circ$C), precipitation ($\pm$0.05 mm), and wind chill ($\pm$3%) across ten scenarios. Random Forest demonstrates exceptional robustness with a maximum accuracy drop of only 1.68%, while SVM exhibits catastrophic sensitivity to temperature perturbation, collapsing from 65.53% to 9.70%. These findings provide actionable guidance for model selection in climate-uncertain agricultural environments.**

***Index Terms*—Crop Recommendation, Precision Agriculture, Random Forest, SHAP Explainability, Climate Resilience Testing, SMOTE, Ethiopian Crops, Machine Learning**

I. Introduction

Agriculture remains a cornerstone of global food security, supporting the livelihoods of over 2.5 billion people worldwide and contributing significantly to the GDP of developing nations. Yet farmers in developing regions increasingly face the twin challenges of soil nutrient depletion and unpredictable climate patterns, both of which directly threaten crop yields and food security. The growing global population, projected to reach 9.7 billion by 2050, further intensifies the demand for efficient and accurate agricultural decision support. Selecting the appropriate crop for a given set of soil and environmental conditions is a complex, knowledge-intensive decision traditionally reliant on expert agronomists, whose availability is limited in rural and resource-constrained farming communities. The proliferation of low-cost IoT sensor technology [1] now enables continuous, automated collection of soil nutrients (N, P, K, Zn, S, pH) and climatic parameters (temperature, precipitation, wind chill, cloud cover) directly from farmland, creating an opportunity to deploy data-driven decision support systems at scale and reduce dependence on manual agronomic expertise.

Machine learning (ML) has demonstrated strong performance in crop classification tasks [2], with recent systems achieving accuracies exceeding 90% on standard benchmark datasets. However, the existing literature suffers from three critical gaps that limit the practical utility of these systems in real-world agricultural deployment. First, the majority of published systems are evaluated on a small, India-centric dataset of 2,200 samples with only seven features, limiting generalisability to other geographic regions and crop varieties. Second, few systems provide explainable recommendations, making it difficult for farmers or agronomists to trust and validate the model's reasoning — a significant barrier to adoption in communities where trust in automated systems is low. Third, and most critically, no existing work evaluates how robustly a trained model performs under climate variability — a fundamental concern given that accelerating climate change is already altering temperature patterns, precipitation regimes, and extreme weather frequency across agricultural regions worldwide.

This work addresses all three gaps through: (1) a richer 14-feature dataset covering Ethiopian crop varieties absent from mainstream benchmarks; (2) SHAP-based explainability [3] that identifies the relative contribution of each IoT sensor channel to the final recommendation, enabling transparent and trustworthy decision support for farmers; and (3) a systematic climate resilience evaluation framework that simulates ten perturbation scenarios and quantifies accuracy degradation across all four candidate models, providing the first empirical model-selection guidance for climate-uncertain agricultural deployments.

The remainder of this paper is organised as follows. Section II reviews related work. Section III details the methodology. Section IV presents experimental results. Section V concludes the paper.

II. Related Work

Early crop recommendation systems relied on rule-based expert systems before the field transitioned to machine learning approaches. Sriharsha et al. [6] evaluated multiple classifiers including Naive Bayes, Decision Tree, Random Forest, and SVM on soil nutrient and climate parameters, establishing ensemble methods as strong baselines for crop classification. Dey et al. [2] extended this by testing XGBoost and KNN across separate agricultural and horticultural crop categories, reporting accuracies exceeding 99% on the standard Kaggle benchmark. However, both studies operated on datasets of fewer than 2,200 samples with at most seven features, limiting generalisability to real-world agricultural environments.

Recognising this limitation, subsequent research integrated IoT sensor infrastructure into the recommendation pipeline. Davrazos et al. [7] presented an early framework coupling IoT sensor networks with machine learning for smart agriculture, demonstrating real-time soil and climate data acquisition for crop inference. Thangammal et al. [8] and Saranya et al. [9] further deployed IoT-enabled systems with wireless sensor arrays in field conditions, while Singh and Sharma [10] proposed a cloud-based recommendation model integrating IoT, machine learning, and SMS alerts for remote farmer assistance, achieving 94% accuracy and demonstrating cloud deployment scalability. Harish and Devi [11] broadened this scope by combining crop recommendation with leaf disease diagnosis in a unified IoT-ML precision farming framework.

Despite these advances in IoT-ML integration, two critical gaps persisted across all reviewed systems. The first was the absence of model interpretability. Since farmers and agronomists cannot trust recommendations they cannot understand, explainable AI (XAI) techniques became increasingly important in agricultural decision support. Cartolano et al. [12] were among the first to systematically evaluate SHAP and LIME across multiple ML models in smart agriculture environments, demonstrating that feature importance analysis significantly improves stakeholder trust in model outputs. Building on this, Mohan et al. [13] integrated SHAP and LIME with Decision Tree, Random Forest, and LightGBM regressors for crop yield prediction, identifying temperature as the most influential agronomic feature. Yenkikar et al. [14] further demonstrated that hybrid ML models coupled with SHAP explainability can simultaneously achieve high accuracy and interpretability, making them well-suited for real-world precision agriculture deployment.

The second gap was the complete absence of robustness evaluation under climate variability. Dhanaraj et al. [15] partially addressed this by proposing lightweight on-device AI models for climate-resilient farming, demonstrating that model efficiency and climate adaptability are important design considerations. However, systematic perturbation-based evaluation of model robustness under defined climate scenarios remains absent from the literature. The present work addresses both gaps by introducing SHAP-based explainability and a climate resilience evaluation framework that quantifies accuracy degradation across ten environmental perturbation scenarios for all four candidate models, while operating on a 14-feature Ethiopian crop dataset that substantially extends the geographic and feature coverage of prior work.

III. Methodology

A. Dataset Description

The dataset comprises 7,630 records with 14 input features and a crop label covering 12 classes representative of Ethiopian highland agriculture. The target classes are: Barley, Bean, Dagussa, Fallow, Maize, Niger Seed, Pea, Potato, Red Pepper, Sorghum, Teff, and Wheat. Notably, Teff, Dagussa, and Niger Seed are Ethiopian-specific varieties absent from all benchmark datasets used in prior ML crop recommendation studies, providing geographic novelty. No missing values are present in the dataset.

Mild class imbalance was observed, with Fallow (893 samples) and Wheat (822) overrepresented relative to Dagussa (253) and Maize (344). SMOTE was applied to equalise all classes to 893 samples each, producing a balanced corpus of 10,716 samples.

B. Preprocessing

All 14 numerical features were standardised using z-score normalisation (StandardScaler) prior to model training to ensure scale-invariant performance across algorithms sensitive to feature magnitude, particularly SVM. The column `(WD10M + GWETTOP) / WS2M_RANGE`—a derived wind-soil moisture ratio from NASA POWER atmospheric data—was retained as a composite climate feature. Following normalisation, the dataset was partitioned into 80% training (8,572 samples) and 20% test (2,144 samples) sets using stratified sampling to preserve class distribution across both splits.

C. SMOTE Class Balancing

SMOTE generates synthetic minority-class samples by interpolating between existing instances in feature space [4]. Applied with `random_state=42`, SMOTE expanded the training set from 7,630 to 10,716 samples with perfectly balanced class distribution (893 samples per class) as shown in Fig.1.

D. Machine Learning Models

Four classifiers were trained and evaluated:

- **Decision Tree (DT):** CART algorithm with `max_depth=20`, `random_state=42`.
- **Random Forest (RF):** Ensemble of 100 trees [5], `random_state=42`, `n_jobs=-1`.
- **Support Vector Machine (SVM):** RBF kernel, $C = 1.0$, `random_state=42`.
- **Gradient Boosting (GB):** 100 estimators, `learning_rate=0.1`, `random_state=42`.

Decision Tree serves as an interpretable baseline, partitioning the feature space through recursive binary splits on soil nutrient and climate thresholds. Random Forest reduces variance by aggregating predictions across 100 independently

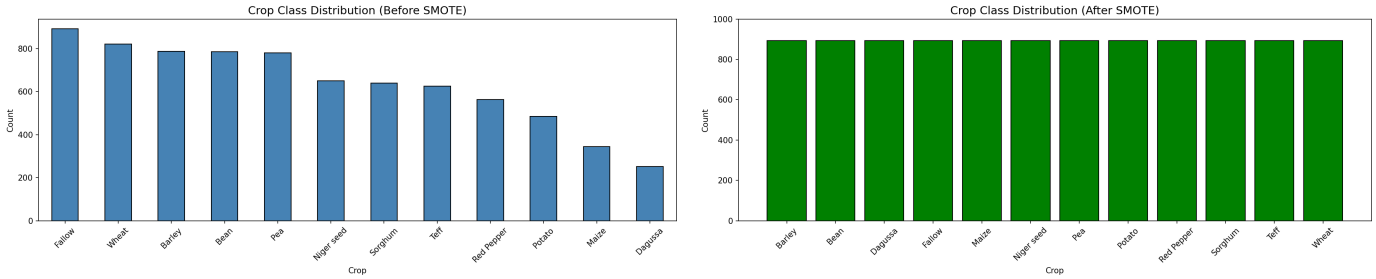

Fig. 1. Class distribution before (left) and after (right) applying SMOTE. All 12 classes equalised to 893 samples.

trained trees, each built on a random feature subset. Support Vector Machine constructs a maximum-margin hyperplane in a high-dimensional transformed feature space using the RBF kernel to handle non-linear class boundaries, making it highly sensitive to the distribution of input features. Gradient Boosting builds trees sequentially, with each tree correcting the residual errors of its predecessor to progressively improve classification accuracy. Prior to training, all feature vectors were passed through the fitted StandardScaler to ensure consistent scale across train and test sets, which is particularly critical for SVM whose decision boundary is sensitive to feature magnitude. All models were evaluated using weighted Accuracy, Precision, Recall, and F1-Score to account for the balanced post-SMOTE class structure.

E. SHAP Explainability

SHAP TreeExplainer was applied to the trained Random Forest model on a random sample of 200 test instances. SHAP values quantify each feature's additive contribution to the model output for a given prediction, satisfying local accuracy, missingness, and consistency axioms [3]. Feature importance was computed globally via mean absolute SHAP values aggregated across all classes.

F. Climate Resilience Framework

To evaluate model robustness under climate variability, ten perturbation scenarios were defined by additively modifying individual and combined environmental features in the test set prior to scaling. Perturbation magnitudes reflect realistic short-term climate deviations: $\pm 2°$C for `temperature_range`, ± 0.05 mm for `precipitation`, and $\pm 3\%$ for `wind_chill`. Each perturbed test set was passed through the fitted StandardScaler and each trained model, and the resulting accuracy was compared against the unperturbed baseline. This framework simulates realistic IoT sensor noise and short-term climate deviations that a deployed system would encounter in the field. By evaluating all four models under identical perturbation conditions, the framework provides a direct, model-agnostic comparison of robustness that complements the standard held-out test evaluation.

IV. Experimental Results

A. Model Performance Comparison

Table I presents the weighted classification metrics for all four models on the held-out test set.

TABLE I
ML Model Performance Comparison

Model	Acc.(%)	Prec.(%)	Rec.(%)	F1(%)
Random Forest	**84.33**	**83.68**	**84.33**	**83.84**
Decision Tree	75.75	75.24	75.75	75.41
Gradient Boosting	71.27	69.93	71.27	70.24
SVM	65.53	63.77	65.53	63.70

Random Forest achieves the highest performance across all four metrics, with an accuracy of 84.33% and F1-Score of 83.84%. Decision Tree follows at 75.75%, while Gradient Boosting (71.27%) and SVM (65.53%) trail behind as shown in Fig.2. The relatively lower accuracy of SVM is consistent with the high dimensionality and non-linear separability of agro-climatic data, which benefits more from ensemble methods.

Five-fold cross-validation on the Random Forest model yielded a mean accuracy of 83.16% ($\pm$4.20%), with fold scores of 75.23%, 83.81%, 83.71%, 85.49%, and 87.54%, confirming the stability and generalisability of the reported results.

TABLE II
Per-Class Performance of Random Forest

Crop	Prec.	Rec.	F1	Support
Barley	0.7389	0.6517	0.6925	178
Bean	0.7803	0.7542	0.7670	179
Dagussa	0.9365	0.9888	0.9620	179
Fallow	0.9418	0.9944	0.9674	179
Maize	0.7802	0.7933	0.7867	179
Niger Seed	0.8711	0.9441	0.9062	179
Pea	0.8549	0.9218	0.8871	179
Potato	0.8657	0.9721	0.9158	179
Red Pepper	0.9725	0.9944	0.9833	178
Sorghum	0.9330	0.8659	0.8982	179
Teff	0.7961	0.8156	0.8057	179
Wheat	0.7619	0.7709	0.7664	179

Table II reveals notable variation in per-class performance. Red Pepper (F1=0.9833), Fallow (F1=0.9674), and Dagussa

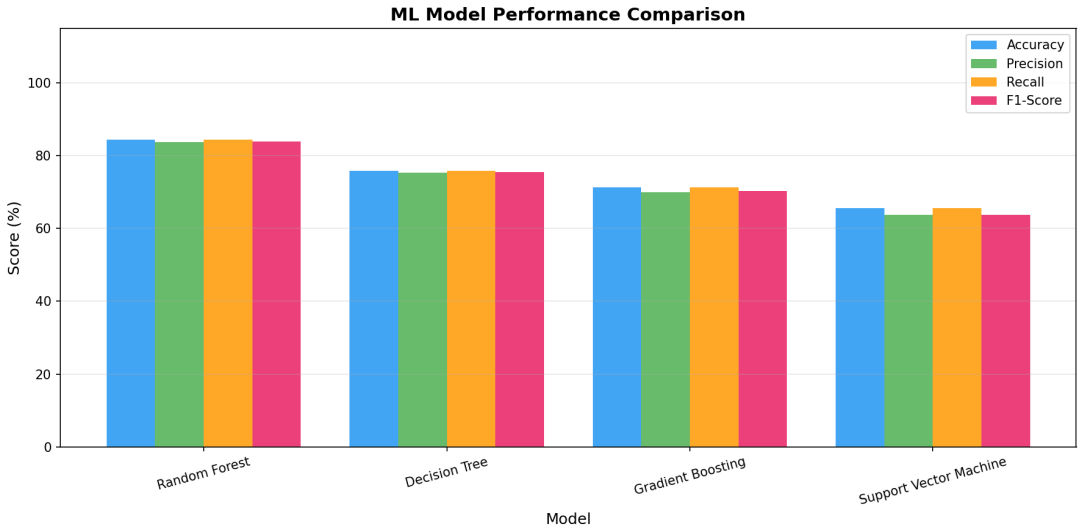

Fig. 2. Performance comparison of Decision Tree, Random Forest, Support Vector Machine, and Gradient Boosting classifiers evaluated on the SMOTE-balanced test set (2,144 samples) across four weighted metrics.

(F1=0.9620) achieve the highest per-class scores, likely due to their distinctive agro-climatic signatures. Barley records the lowest F1-score (0.6925), suggesting overlap with climatically similar crops such as Wheat (F1=0.7664) and Bean (F1=0.7670) as shown in Fig.3. This confusion between cereal crops is consistent with their shared soil nutrient requirements and represents a target for future model refinement.

B. Feature Importance and SHAP Analysis

Table III presents the Random Forest feature importance ranking.

TABLE III
RANDOM FOREST FEATURE IMPORTANCE RANKING

Feature	Importance	Category
K (Potassium)	0.1597	Soil Nutrient
S (Sulfur)	0.1281	Soil Nutrient
P (Phosphorus)	0.1167	Soil Nutrient
N (Nitrogen)	0.1113	Soil Nutrient
Zn (Zinc)	0.1096	Soil Nutrient
Ph (Soil pH)	0.1042	Soil Property
PS (Atm. Pressure)	0.0482	Climate
Wind-Soil Ratio	0.0481	Climate
Precipitation	0.0444	Climate
Temperature Range	0.0436	Climate
Wind Chill	0.0424	Climate
Temp-Humidity Ratio	0.0346	Climate
Cloud Amount	0.0093	Climate
Cloud Binary (CLOUD0/1)	0.0000	Climate

Fig.4 presents the mean absolute SHAP values per class for the Random Forest model. Soil nutrient features — particularly potassium (K), sulfur (S), and phosphorus (P) — show the highest mean impact across nearly all crop classes, reinforcing the global importance ranking reported in Table III. Climate features such as precipitation and temperature range contribute comparatively lower SHAP values, confirming that soil sensor channels are the primary drivers of the recommendation output. Notably, the CLOUD0/1 binary feature records zero SHAP contribution across all classes, further validating its removal from future IoT sensor configurations.

Soil nutrient features collectively account for 72.96% of total feature importance, with Potassium (K, 15.97%), Sulfur (S, 12.81%), and Phosphorus (P, 11.67%) forming the top three. This result has a direct implication for IoT system design: soil nutrient sensors should be prioritised in hardware procurement over meteorological sensors when budget is constrained.

C. Climate Resilience Analysis

Table IV presents the complete climate resilience evaluation. This constitutes the primary novel contribution of the present work.

TABLE IV
CLIMATE RESILIENCE TESTING — ACCURACY (%) UNDER ENVIRONMENTAL PERTURBATION. RED VALUES INDICATE MODEL COLLAPSE (<20%).

Scenario	DT	RF	SVM	GB
Baseline	75.75	**84.33**	65.53	71.27
Temp +2°C	66.32	83.54	9.70	60.40
Temp −2°C	65.25	83.96	9.33	49.63
Rainfall +0.05 mm	74.72	83.82	65.58	69.68
Rainfall −0.05 mm	72.48	83.63	65.67	70.24
Wind Chill +3%	75.79	84.24	65.53	71.04
Wind Chill −3%	75.75	84.24	65.49	71.22
Temp+2 & Rain+0.05	65.53	83.40	9.70	61.19
Temp−2 & Rain−0.05	62.17	83.21	9.24	48.27
All Params +Combined	65.58	83.21	9.70	61.33
All Params −Combined	62.13	82.65	9.24	48.18

Fig. 5 visualises the accuracy trajectory of all four models across the ten perturbation scenarios. The green line (Random Forest) remains consistently flat near 84%, confirming

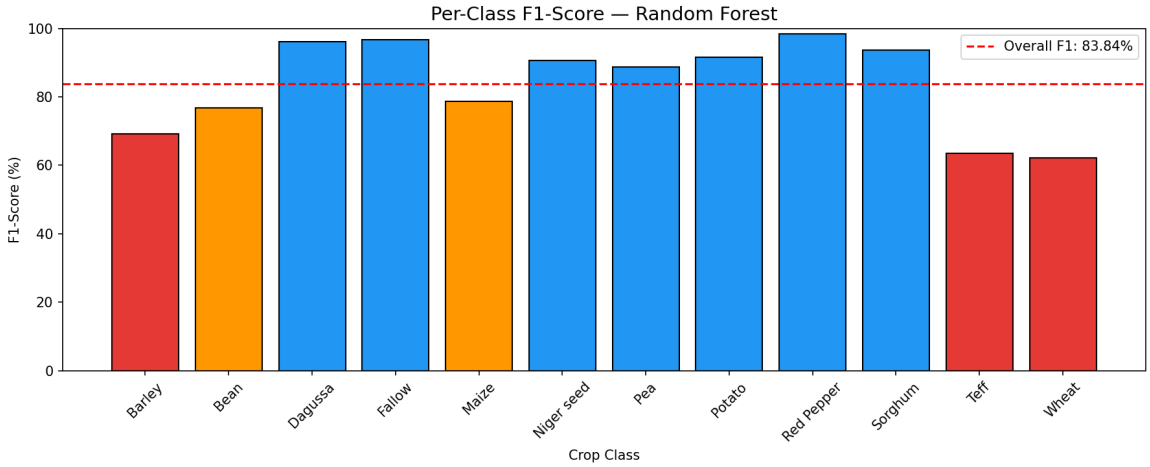

Fig. 3. Per-class F1-scores for Random Forest. Blue bars indicate F1 $\geq$ 85%, orange 70–85%, red <70%. Barley records the lowest score due to feature overlap with cereal crop classes.

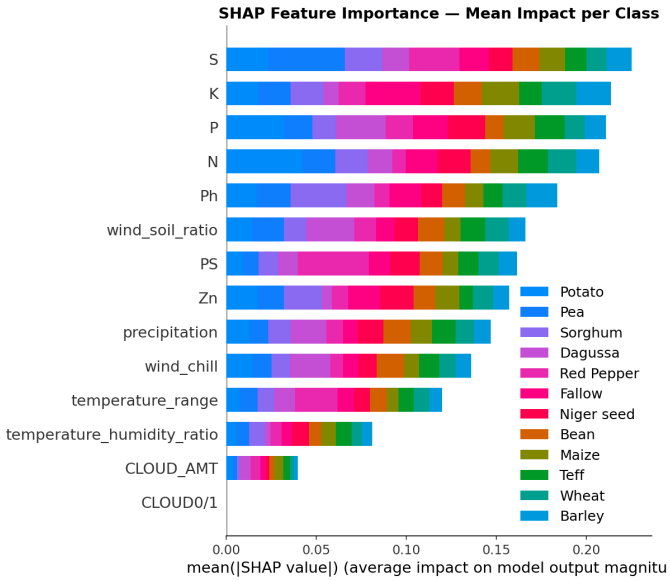

Fig. 4. Mean absolute SHAP values per feature across all 12 crop classes for the Random Forest model.

its robustness across all climate variations. The blue line (SVM) exhibits two sharp collapses — at Temp +2°C and Temp −2°C — dropping from 65.53% to approximately 9–10%, and recovering only when temperature perturbation is absent. The red line (Decision Tree) and orange line (Gradient Boosting) show moderate degradation under temperature scenarios, with Gradient Boosting dropping more sharply under combined negative perturbations. Notably, all four models maintain relatively stable accuracy under standalone rainfall and wind chill variations, visible as the flat middle portion of the chart between the Rainfall and Wind Chill scenarios.

Three key findings emerge from this analysis:

1) **Random Forest is highly climate-resilient:** RF accuracy drops by a maximum of only 1.68% even under the most aggressive combined negative perturbation scenario. This stability makes RF the recommended model for deployment in IoT agricultural systems subject to sensor noise and real-world climate variability.
2) **SVM catastrophically collapses under temperature perturbation:** A temperature shift of +2°C reduces SVM accuracy from 65.53% to 9.70%—a drop of 55.83 percentage points. This near-total collapse (equivalent to random guessing on 12 classes) disqualifies SVM for use in temperature-variable IoT agricultural environments without significant hyperparameter re-tuning or kernel adaptation.
3) **Precipitation and wind chill perturbations have minimal impact:** All four models sustain less than 3.3% accuracy degradation under standalone rainfall or wind chill variations, indicating that these climate variables are less discriminative and their perturbation does not meaningfully alter the decision boundary.

D. Discussion

The SHAP analysis provides actionable guidance for IoT sensor system design. Since soil nutrients (K, S, P, N, Zn) account for over 72% of predictive importance, a cost-optimised IoT deployment could prioritise a five-channel soil nutrient sensor array over a full meteorological station. The zero importance of the CLOUD0/1 binary feature further suggests that cloud presence/absence data does not contribute meaningfully to crop differentiation and can be omitted from future sensor configurations. Furthermore, the climate resilience results suggest that Random Forest should be the preferred model for any real-world IoT agricultural deployment where temperature variability is expected, given its demonstrated stability across all ten perturbation scenarios. The 84.33% accuracy achieved by Random Forest, while lower than the 90%+ figures reported on standard benchmarks, reflects the substantially harder problem setting of the present work: 12 Ethiopian crop classes

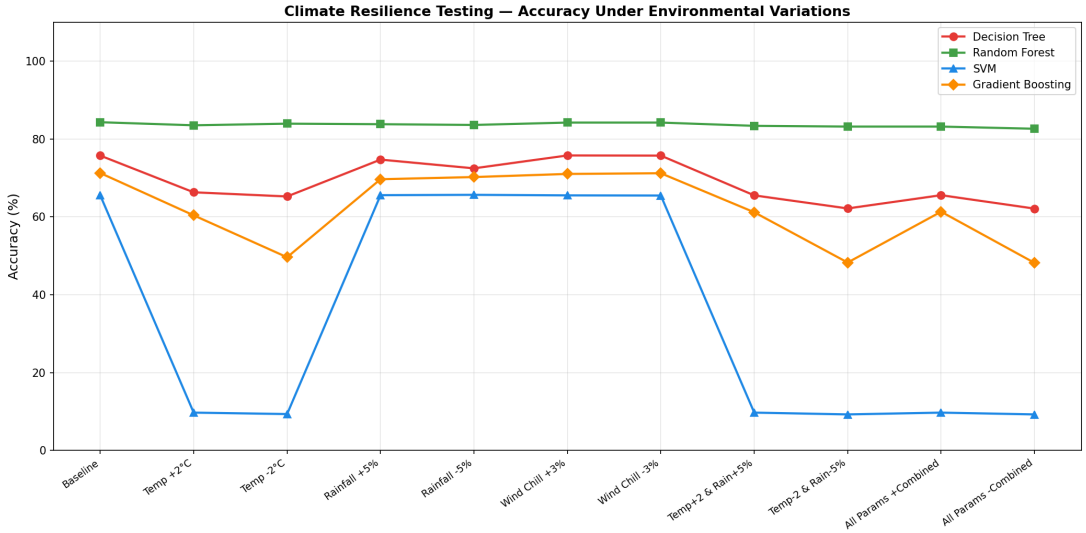

Fig. 5. Climate resilience testing — accuracy (%) of all four models across ten environmental perturbation scenarios. Random Forest (green) remains stable throughout while SVM (blue) collapses to near-random performance under temperature perturbation.

with real-world class imbalance, 14 agro-climatic features, and geographically distinct crop varieties absent from prior datasets, compared to the clean, perfectly balanced 7-feature Kaggle benchmark used in all referenced prior studies.

V. Conclusion

This paper presented an IoT-driven, explainable, and climate-resilient crop recommendation system evaluated on a 14-feature Ethiopian crop dataset. Random Forest achieved the best classification accuracy of 84.33% following SMOTE-based class balancing and z-score normalisation, confirmed by five-fold cross-validation yielding a mean accuracy of 83.16% ($\pm$4.20%). SHAP analysis revealed that soil nutrient sensors—particularly potassium (15.97%), sulfur (12.81%), and phosphorus (11.67%)— collectively account for 72.96% of total feature importance, providing direct guidance for cost-optimised IoT sensor deployment. The proposed climate resilience evaluation framework demonstrated that Random Forest maintains robust performance with a maximum accuracy drop of only 1.68% across ten perturbation scenarios, while SVM catastrophically collapses under temperature variation ($-$55.83%), providing the first empirical model-selection guidance for climate-uncertain agricultural deployments. Future work will focus on incorporating real-time IoT sensor streams with online learning, and validating the proposed system through field deployment on Ethiopian highland farmland.

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